

Cognitive Computing

Introduction to Natural language processing (NLP)

Feature Extraction from Text

- Unstructured data cannot be represented in tabular format.
 - Therefore, it is essential to convert it to numeric features because most machine learning algorithms are capable of dealing only with numbers
- Features - General & Specific
- Two Popular Methods of Feature (Special) Extraction from Text:
 - Bag of Words (BoW)
 - Term Frequency-Inverse Document Frequency (TF-IDF)

Bag of Words (BoW)

- Bag of Words (BoW) model is one of the most popular methods for extracting features from raw texts.
- Output of this model for a set of text documents is a **MATRIX**
- Each column of the matrix represents a word from the vocabulary and each row corresponds to one of these text document
- *Vocabulary – refers to unique set of words present in a document*

Document 1: I like detective Byomkesh Bakshi.

Document 2: Byomkesh Bakshi is not a detective, he is a truth seeker.

The corresponding BoW representation would be as follows:

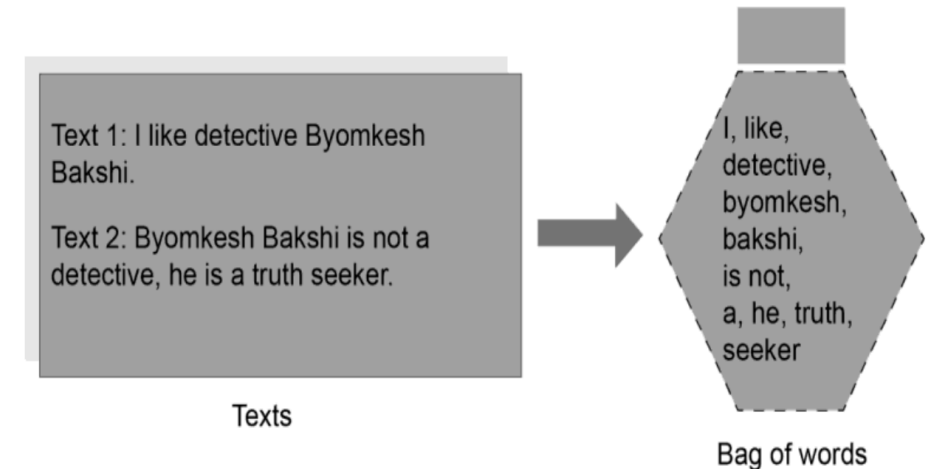


Figure 2.35: Diagram of the BoW model

The tabular representation of the BoW model would be as follows:

	i	like	detective	byomkesh	bakshi	is	not	a	he	truth	seeker
Text-1	1	1	1	1	1	0	0	0	0	0	0
Text-2	0	0	1	1	1	2	1	2	1	1	1

Figure 2.36: Tabular representation of the BoW model

Example

1. 'Data Science is an overlap between Arts and Science',
2. 'Generally, Arts graduates are right-brained and Science graduates are left-brained',
3. 'Excelling in both Arts and Science at a time becomes difficult',
4. 'Natural Language Processing is a part of Data Science'

Question: Create a BoW for the 10 most frequent items

Example

1. 'Data Science is an overlap between Arts and Science',
2. 'Generally, Arts graduates are right-brained and Science graduates are left-brained',
3. 'Excelling in both Arts and Science at a time becomes difficult',
4. 'Natural Language Processing is a part of Data Science'

Most frequent items: an, and, are, arts, brained, data, graduates, is, right, science

	an	and	are	arts	brained	data	graduates	is	right	science
0	1	1	0	1	0	1	0	1	0	2
1	0	1	2	1	2	0	2	0	1	1
2	0	1	0	1	0	0	0	0	0	1
3	0	0	0	0	0	1	0	1	0	1

Bag of Words (BoW)

Limitations:

- Does not consider word importance.
- Common words (e.g., "the", "is", "a") get high weights, even if they don't carry useful meaning.
- It does not handle **semantic meaning** or **context**.

Term Frequency-Inverse Document Frequency (TF-IDF)

- TF-IDF is a statistical measure used in natural language processing and information retrieval to evaluate the importance of a word in a document relative to a collection of documents (corpus).
- Improves BoW by assigning **weights** to words based on their importance.
- Unlike simple word frequency, TF-IDF balances common and rare words to highlight the most meaningful terms.

Term Frequency-Inverse Document Frequency (TF-IDF)

- TF-IDF combines two components:
 - **Term Frequency (TF)** and
 - **Inverse Document Frequency (IDF)**
- **Term Frequency (TF):** Measures how often a word appears in a document. A higher frequency suggests greater importance. If a term appears frequently in a document, it is likely relevant to the document's content.
- **Formula:**

$$TF(t, d) = \frac{\text{Number of times term } t \text{ appears in document } d}{\text{Total number of terms in document } d}$$

Term Frequency-Inverse Document Frequency (TF-IDF)

- Inverse Document Frequency (IDF): Reduces the weight of common words across multiple documents while increasing the weight of rare words. If a term appears in fewer documents, it is more likely to be meaningful and specific.
- Formula:

$$\text{IDF}(t, D) = \ln \frac{\text{Total number of documents in corpus } D}{\text{Number of documents containing term } t}$$

Example-1

- Imagine we have a corpus (a collection of documents) with three documents:
 - Document 1: The cat sat on the mat
 - Document 2: The dog played in the park
 - Document 3: Cats and dogs are great pets

Our goal is to calculate the TF-IDF score for specific terms in these documents. Let's focus on the word "cat" and see how TF-IDF evaluates its importance.

$$\text{TF}(\text{cat}, 1) = 1/6. \quad \text{IDF}(\text{cat}, D) = \ln(3/2) = 0.54$$

$$\text{TF}(\text{cat}, 2) = 0/6$$

$$\text{TF}(\text{cat}, 3) = 1/6$$

Example-1:Solution

- Step 1: Calculate Term Frequency (TF)
- For Document 1:
 - The word “cat” appears 1 time.
 - The total number of terms in Document 1 is 6 (“the”, “cat”, “sat”, “on”, “the”, “mat”).
 - So, $TF(\text{cat}, \text{Document 1}) = 1/6$
- For Document 2:
 - The word “cat” does not appear.
 - So, $TF(\text{cat}, \text{Document 2})=0$.
- For Document 3:
 - The word “cat” appears 1 time (as “cats”).
 - The total number of terms in Document 3 is 6 (“cats”, “and”, “dogs”, “are”, “great”, “pets”).
 - So, $TF(\text{cat}, \text{Document 3})=1/6$

Example-1:Solution

- Step 1: Calculate Term Frequency (TF)
 - In Document 1 and Document 3, the word “cat” has the same TF score. This means it appears with the same relative frequency in both documents.
 - In Document 2, the TF score is 0 because the word “cat” does not appear.

Example-1:Solution

- Step-2: Calculate Inverse Document Frequency (IDF)
- Total number of documents in the corpus (D): 3
- Number of documents containing the term “cat”: 2 (Document 1 and Document 3).
- So, $IDF(cat, D) = \ln 3 / 2 = 0.54$
- The IDF score for “cat” is relatively low.
- This indicates that the word “cat” is not very rare in the corpus—it appears in 2 out of 3 documents. If a term appeared in only 1 document, its IDF score would be higher, indicating greater uniqueness.

Example-1:Solution

- Step 3: Calculate TF-IDF
 - The TF-IDF score for “cat” is 0.09 in Document 1 and Document 3, and 0 in Document 2 that reflects both the frequency of the term in the document (TF) and its rarity across the corpus (IDF).

The TF-IDF score is the product of TF and IDF:

$$\mathbf{TF-IDF(t, d, D) = TF(t, d) \times IDF(t, D)}$$

For Document 1:

$$\mathbf{TF-IDF(cat, Document\ 1, D) = 0.167 \times 0.54 \approx 0.090}$$

For Document 2:

$$\mathbf{TF-IDF(cat, Document\ 2, D) = 0 \times 0.54 = 0}$$

For Document 3:

$$\mathbf{TF-IDF(cat, Document\ 3, D) = 0.167 \times 0.54 \approx 0.090}$$

- A higher TF-IDF score means the term is more important in that specific document.

Example-1:Solution

- **Why is TF-IDF Useful in This Example?**

- 1. Identifying Important Terms:**

- TF-IDF helps us understand that “cat” is somewhat important in Document 1 and Document 3 but irrelevant in Document 2.

If we were building a search engine, this score would help rank Document 1 and Document 3 higher for a query like “cat”.

- 2. Filtering Common Words:**

- Words like “the” or “and” would have high TF scores but very low IDF scores because they appear in almost all documents.
- Their TF-IDF scores would be close to 0, indicating they are not meaningful.

Example-1:Solution

3. Highlighting Unique Terms:

- If a term like “mat” appeared only in Document 1, it would have a higher IDF score, making its TF-IDF score more significant in that document.

Example-2

- Imagine we have a corpus (a collection of documents) with two documents:
- **Document A: The Car Is Driven On the Road**
- **Document B: The Truck is Driven on the highway**

Calculate the TF-IDF score for each term.

$$TF(\text{driven}, 1) = 1/7$$

$$TF(\text{driven}, 2) = 1/7$$

$$IDF(\text{driven}, D) = \ln(2/2) = \ln 1 = 0$$

$$Tf-idf = 1/7 * 0 = 0$$

Example-2- Solution:

Word	TF		IDF	TF*IDF	
	A	B		A	B
The	2/7	2/7	$\ln(2/2) = 0$	0	0
Car	1/7	0	$\ln(2/1) = 0.693$	0.099	0
Truck	0	1/7	$\ln(2/1) = 0.693$	0	0.099
Is	1/7	1/7	$\ln(2/2) = 0$	0	0
Driven	1/7	1/7	$\ln(2/2) = 0$	0	0
On	1/7	1/7	$\ln(2/2) = 0$	0	0
Road	1/7	0	$\ln(2/1) = 0.693$	0.099	0
Highway	0	1/7	$\ln(2/1) = 0.693$	0	0.099

Zipf's Law

- Zipf's Law states that in a large corpus of natural language text, the frequency of a word is inversely proportional to its rank in the frequency table.

- Mathematically, $f \propto \frac{1}{r}$

Or , $f = \frac{C}{r}$

- Example Interpretation:
 - The most frequent word appears twice as often as the second most frequent word.
 - The third most frequent word appears one-third as often as the most frequent word.

Why Zipf's Law is Useful?

- Explains Word Frequency Distribution:
 - Students often wonder why some words dominate text datasets.
 - Zipf's Law helps them understand that most natural languages follow a power-law distribution, meaning a few words appear very frequently, while many words appear rarely.
- Motivates the Need for TF-IDF:
 - In BoW, high-frequency words (like "the," "is," "and") dominate, leading to poor differentiation between documents.
 - Zipf's Law helps justify why we need IDF (Inverse Document Frequency) to balance word importance in TF-IDF.
- Justifies Stopword Removal:
 - Since Zipf's Law shows that high-frequency words carry little meaning, it provides a theoretical basis for removing stop words in NLP.

Phase 2: Syntactic Analysis

2.1 Syntax - PoS Tagging, Sentence Structure

2.2 Parsing – Constituency and Dependency

Step 2. Syntax and Parsing

- How sentences are structured and how words connect to form meaning
 - Syntax defines the grammar rules,
 - Parsing analyzes sentences and converts them into structured representations like parse trees, helping machines understand relationships between words.
- Understanding the structure of a sentence and assessing its grammatical correctness.
- Analyzing the relationships between words and ensuring their logical consistency by comparing their arrangement against standard grammatical rules.

Step 2.1 Syntax

- **Parts of Speech(PoS) Tagging**

- PoS Tagging - Identifying nouns, verbs, adjectives, etc.



- **Sentence structure**

Common pattern is subject + verb + object
Defines how information flows in a sentence

Sentence: *The cat eats fish*

- **Subject** → The cat
- **Verb** → eats
- **Object** → fish

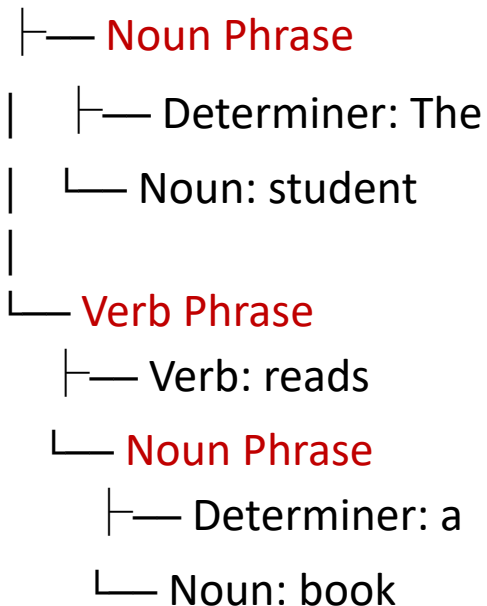
This structure tells the model:

- who performs the action
- what action is happening
- what receives the action

Step 2.2 Parsing - Analyzing the grammatical structure of a sentence according to a syntactic representation.

- *Syntax gives us the rules; parsing applies those rules to a particular sentence.*
- *E.g: The student reads a book*
- *A parser determines the following structure – A parse tree*

Sentence



A Noun Phrase (NP) is a group of words that functions together as a noun in a sentence

- *The dog*
- *The tall man*
- *The cute dog on the sofa*

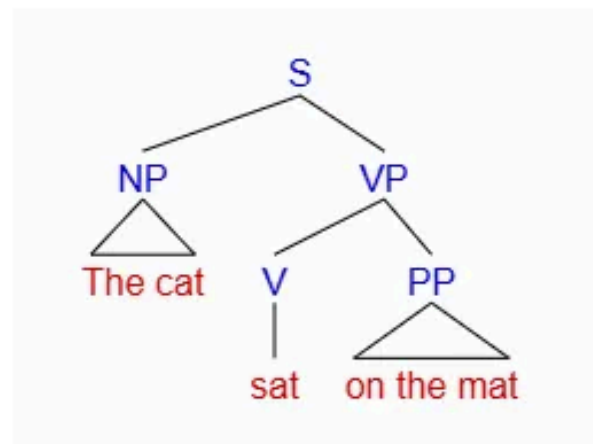
A Verb Phrase (VP) a group of words consisting of a main verb and any auxiliary (helping) verbs, modifiers, or objects

- *She is reading a book. (is = helping verb, reading = main verb)*
- *They will arrive tomorrow. (will = modal verb, arrive = main verb)*
- *He should have been working*

Step 2.2 Parsing – Parse Tree

“The cat sat on the mat.”

- Noun Phrase: “The cat”
- Verb Phrase: “sat on the mat”
- Within the verb phrase, “on the mat” is a prepositional phrase.

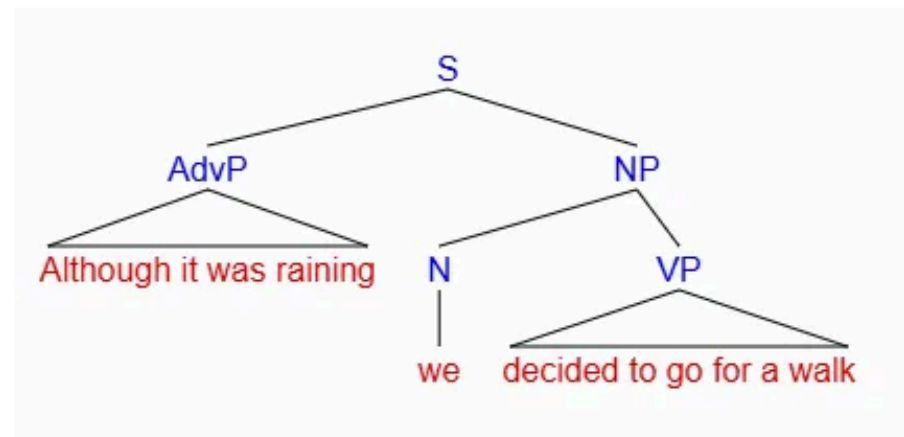


Syntax tree: S= Sentence, NP = Noun Phrase, VP = Verb Phrase, V = Verb, PP = Prepositional Phrase.

Step 2.2 Parsing – Parse Tree

“Although it was raining, we decided to go for a walk.”

- Noun Phrase: “we”
- Verb Phrase: “decided to go for a walk”
- The clause “Although it was raining” is an adverbial phrase modifying the main clause (*we decided to go for a walk*).



Syntax tree: S= Sentence, AdvP = Adverb Phrase, NP = Noun Phrase, VP = Verb Phrase.

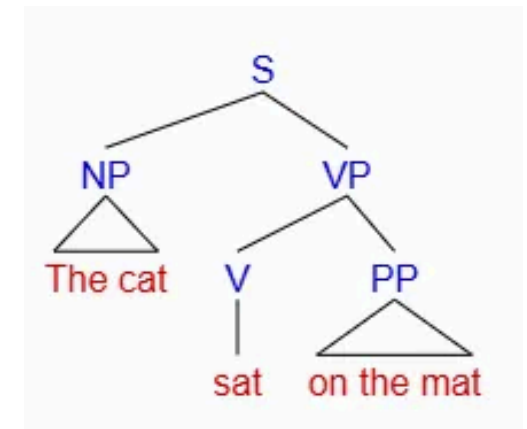
Step 2.2 Parsing

- **Constituency Parsing:**

- Breaking down a sentence into its constituent parts or phrases (e.g., noun phrases, verb phrases).
 - "The cat" → (Determiner + Noun)
 - "A tall building" → (Article + Adjective + Noun)
 - "My best friend" → (Possessive Pronoun + Adjective + Noun)
 - "is running fast" → (Auxiliary Verb + Verb + Adverb)

Example Output

(S (NP The cat) (VP sat (PP on (NP the mat))))



- **Another example: She reads a book**
 - (S(NP She (VP reads (NP (Det)a (N) book))))

Step 2.2 Parsing

- We can define Grammar rules now as:

$S \rightarrow NP VP$

$NP \rightarrow Det N$

$VP \rightarrow V NP$

S

$\rightarrow NP VP$

$\rightarrow Det N VP$

$\rightarrow Det N V NP$

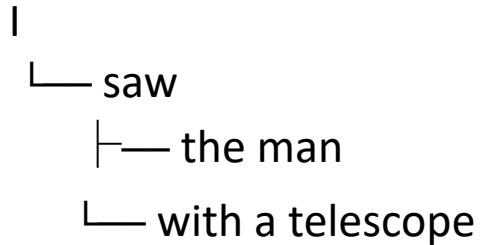
$\rightarrow Det N V Det N$

Why do we need parsing?

- Eg: I saw the man with a telescope.

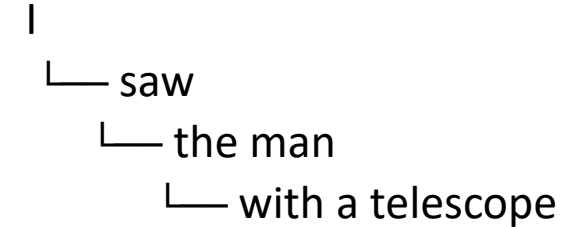
Interpretation 1

I used a telescope to see the man.



Interpretation 2

The man had a telescope.



Same words. Different structure. Different meaning.

But tree exists for both -> AMBIGUITY

Parsing alone does not always resolve meaning.

Parsing provides possible structural interpretations. To decide which interpretation makes sense, an intelligent system may need:

Context, semantic knowledge , world knowledge , previous conversation , probabilities

Step 2.2 Parsing

- **Dependency Parsing:**

- Analyzing the grammatical structure of a sentence to identify relationships between words (word by word).

"The cat sat on the mat."

Example Output

"sat" → (root), "cat" → (subject of sat), "on"
→ (relation to sat), etc.

- **She reads a book.**
- reads → she (subject)
reads → book (object)

- *Dependency parsing shows how words depend on each other*
- *Constituency parsing shows how words group into phrases*

Syntax vs Parsing in NLP

- Syntax is static → it defines rules
- Parsing is dynamic → it applies those rules
- Syntax alone cannot interpret text
- Parsing uses syntax to generate meaning

Aspect	Syntax	Parsing
Meaning	Defines grammar rules	Analyzes sentences using those rules
Role	Sets structure and order	Applies structure to real text
Focus	How sentences should be formed	How sentences are interpreted
Output	Rules for valid sentences	Structured output like parse trees

Constituency vs Dependency Parsing

- DIY

Step 3. Semantic Analysis

- It is the third phase of Natural Language Processing (NLP), focusing on extracting the meaning from text.
- Unlike syntax analysis, which deals with grammatical structure, semantic analysis is concerned with the literal and contextual meaning of words, phrases, and sentences.

Examples of Semantic Analysis

- Consider the following examples: "Apple eats a John."
- Syntactically Correct but Semantically Incorrect.
- This sentence is grammatically correct but does not make sense semantically. An apple cannot eat a person, highlighting the importance of semantic analysis in ensuring logical coherence.

Step 3. Semantic Analysis

Key Tasks in Semantic Analysis:

- Named Entity Recognition (NER)
 - Identify names, locations and dates
- Word Sense Disambiguation (WSD):
 - Determining which meaning of a word is used in a given context.
 - WSD determines the correct meaning of ambiguous words based on context.
 - For example, the word "bank" can refer to a financial institution or the side of a river.
 - WSD uses *contextual clues* to assign the appropriate meaning.

Step 4. Discourse Integration

- **Discourse Integration** in **NLP** refers to the process of understanding how the meaning of a sentence is influenced by the context of the surrounding discourse (previous and following sentences).
- It helps NLP models maintain coherence and continuity in text understanding by considering **prior context, pronouns, references, and logical connections** between sentences.
- Many NLP tasks require understanding text beyond individual sentences and here the Discourse Integration.

Step 4. Discourse Integration

Examples of Discourse Integration

- **Coreference Resolution** - Task of identifying when **two or more expressions** in a text **refer to the same entity**.
 - *"Alice bought a book. She loved it."*
 - Without discourse integration, **"She"** and **"it"** would be ambiguous. NLP models must resolve that **"She" = Alice** and **"it" = book**.
- **Ellipsis Resolution**- Refers to the omission of words in a sentence that are understood from context.
 - *"John went to the park. Mary did too."*
 - What did Mary do? Without discourse understanding, the sentence is incomplete.

Step 4. Discourse Integration

Examples of Discourse Integration

- **Anaphora** - Refers to a word or phrase (typically a pronoun) that refers **backward** to a previously mentioned entity
- **Cataphora Resolution** - Reference is made **before** the entity is introduced (i.e., **forward reference**)
 - **Anaphora (refers back)** → *"I met a dog yesterday. It was very friendly."*
 - **Cataphora (refers forward)** → *"Although he was late, John still managed to finish the task."*
- **Coherence in Dialogue Systems**
 - *User:* "What is the weather today?"
 - *Bot:* "It's sunny. Do you want to know the forecast for tomorrow?"
 - The bot **understands the context** and **extends the conversation naturally**.

Step 5. Pragmatics Analysis

- The fifth and final phase
- It focusing on interpreting the inferred meaning of a text beyond its literal content.
- Human language is often complex and layered with underlying assumptions, implications, and intentions that go beyond straightforward interpretation. This phase aims to grasp these deeper meanings in communication.

Step 5. Pragmatics Analysis

- Role of **Pragmatics Analysis**:
 - Pragmatic analysis goes beyond the literal meanings examined in semantic analysis, aiming to understand what the writer or **speaker truly intends to convey**.
 - In natural language, words and phrases can carry different meanings depending on context, tone, and the situation in which they are used.

Step 5. Pragmatics Analysis

- **Examples:**
- Contextual Greeting: "Hello! What time is it?"
 - "Hello!" is more than just a greeting; it serves to establish contact.
 - "What time is it?" might be a straightforward request for the current time, but it could also imply concern about being late.
- Figurative Expression: "She has a heart of gold."
 - Literal meaning: A heart made of the metal gold.
 - Figurative meaning: The person is very kind and generous.